Solutions for final exam, PCS 125, 2013.

1. Read carefully the following 4 statements. Which are true?

√ **A)** The work done by an external agent to separate two masses m_1, m_2 which are at a distance r to an infinite distance from each other is: $G\frac{m_1 m_2}{r}$.

X **B)** The work done by an external agent to separate two masses m_1, m_2 which are at a distance r to an infinite distance from each other is: $-G\frac{m_1 m_2}{r}$.

X **C)** The work done by the electric field to separate two charges of equal sign q_1, q_2 which are at a distance r to an infinite distance from each other is: $-k_e\frac{q_1 q_2}{r}$.

√ **D)** The work done by the electric field to separate two charges of equal sign q_1, q_2 which are at a distance r to an infinite distance from each other is: $k_e\frac{q_1 q_2}{r}$.

a. A only **b.** B only **c.** A and C **d.** B and D **e.** none of the above

A and D are true

- For gravitational force:

$$W_{ext.} = \Delta U = (U_f - U_i) = (U_\infty - U_r) = \left(0 - \left(-G\frac{m_1 m_2}{r}\right)\right) =$$

$$= G\frac{m_1 m_2}{r}$$

Since the force acting between two masses is attractive, the work done by external agent to seperate two masses located at the distance r to an infinite distance from each other is positive.

- For electric force:

$$W_{int} = -\Delta U = -(U_f - U_i) = -(U_\infty - U_r) = -\left(0 - k_e\frac{q_1 q_2}{r}\right) =$$

$$= +k_e\frac{q_1 q_2}{r} \quad (q_1 > 0,\ q_2 > 0 \text{ or } q_1 < 0,\ q_2 < 0 \Rightarrow q_1 q_2 > 0)$$

Since the force acting between two charges of the same sign is repulsive, the work done by electric field to seperate two charges of the same sign located at distance r to an infinite distance from each other is positive.

2. An object of mass m is released from rest at a distance h above the surface of a planet (with mass = M, radius = R, where $R < h$). With what speed will the object strike the surface of the planet? Disregard any dissipative effects of the atmosphere of the planet.

a. $\left[\frac{2GMh}{R(R+h)}\right]^{1/2}$ b. $\left[\frac{2GM}{R}\right]^{1/2}$ c. $\left[\frac{2GM(h-R)}{Rh}\right]^{1/2}$

d. $\left[\frac{2GM}{R+h}\right]^{1/2}$ e. $\left[\frac{2GMh}{R+h}\right]^{1/2}$

$E_i = E_f$ (total energy is conserved)

$E_i = K_i + U_i = \frac{mv_i^2}{2} - G\frac{mM}{(R+h)} = 0 - G\frac{mM}{(R+h)}$

since object is released from rest its initial speed is 0

R = radius of a planet

M = mass of a planet

m = mass of object

$E_f = K_f + U_f = \frac{mv_f^2}{2} - \frac{GmM}{R}$

$E_i = E_f \Rightarrow -\frac{G\cancel{m}M}{R+h} = \frac{\cancel{m}v_f^2}{2} - \frac{G\cancel{m}M}{R} \Rightarrow$

$v_f = \sqrt{2\left[\frac{GM}{R} - \frac{GM}{R+h}\right]}$ $\sqrt{2GM\frac{\cancel{R}+h-\cancel{R}}{R(R+h)}} = \sqrt{\frac{2GMh}{R(R+h)}}$

3. The oscillation of a 2.0-kg mass on a spring is described by $x(t) = 3.0\cos(4.0t + 0.8)$ where x is in centimeters and t is in seconds. What is the force constant k of the spring?

a. 4.0 N/m
b. 0.80 N/m
c. 16 N/m
d. 32 N/m
e. 2.0π N/m

$x(t) = 3.0\cos(4.0t + 0.8)$

$\omega = 4.0 = \sqrt{\frac{k}{m}}$

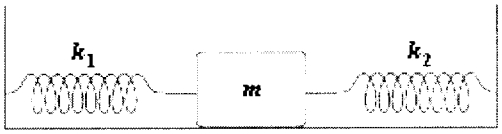
$k = \omega^2 m = (4.0\ s^{-1})^2(2.0\ kg) =$

$= 32\ \frac{kg}{s^2} = 32\ \frac{kg \cdot m}{s^2 \cdot m} = 32\ \frac{N}{m}$

When the mass is displaced a distance x from equilibrium, spring 1 is stretched a distance x and spring 2 is compressed a distance x: $x = A$

4. Observe the figure given in this question. The block of mass m slides on a frictionless surface attached as shown to two springs with constants $k_1 = 400$ N/m and $k_2 = 600$ N/m.
If the mass m is initially displaced to the right 3.0 cm, what is the maximum kinetic energy of the mass m?

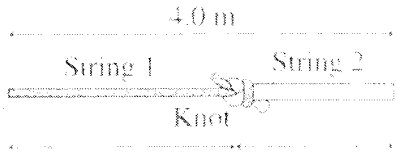
Force exerted on m is directed to the left an is $F = -k_1x - k_2x = -(k_1+k_2)x$ in magnitude.



a. 10 J
b. 15 J
c. 0.45 J
d. 0.94 J
e. cannot calculated because m is not given

$$K_{max} = U_{max} = \frac{1}{2}kA^2 = \frac{1}{2}(k_1+k_2)A^2 = \frac{1}{2}\left(400\frac{N}{m} + 600\frac{N}{m}\right)(0.03m)^2 = 0.45J$$

5. String 1 in the figure below has linear density 2.0 g/m and string 2 has linear density 4.0 g/m. A student sends pulses in both directions by quickly pulling up on the knot, then releasing it. What should be the string lengths L_1 and L_2 if the pulses are to reach the ends of the strings simultaneously?



a. L_1=2.34m, L_2=1.66m
b. L_1=2.36m, L_2=1.64m
c. L_1=2.38m, L_2=1.62m
d. L_1=2.40m, L_2=1.60m
e. L_1=2.42m, L_2=1.58m

$$t_1 = \frac{L_1}{\sqrt{\frac{T}{\mu_1}}} \quad \Big| \quad t_2 = \frac{L_2}{\sqrt{\frac{T}{\mu_2}}}$$

$$t_1 = t_2 \Rightarrow L_1\sqrt{\mu_1} = L_2\sqrt{\mu_2} \Rightarrow$$

$$\frac{L_1}{L_2} = \sqrt{\frac{\mu_2}{\mu_1}} = \sqrt{\frac{4.0\ g/m}{2.0\ g/m}} = \sqrt{2} \quad \Big| \quad L_1 + L_2 = 4m \quad L_1 = \sqrt{2}L_2 \quad \Big| \quad (\sqrt{2}+1)L_2 = 4m \quad L_2 = \frac{4m}{\sqrt{2}+1} = 1.66m \quad L_1 = 2.34m$$

6. A spherical wave travels outward from the origin. At a particular instant, the phase of the wave at a point 20 cm distant from the source (r_1=20 cm) is 0 rad and the phase at a point 80 cm from the source (r_2=80 cm) is 3π rad. What is the wavelength of the wave?

a. 20cm
b. 30cm
c. 40cm
d. 50cm
e. Can not be determined because time is unknown

$$\Delta\varphi = k\Delta r = \frac{2\pi}{\lambda}\Delta r = 3\pi \Rightarrow \lambda = \frac{2}{3}\Delta r = \frac{2}{3}(\overset{r_2}{80\,cm} - \overset{r_1}{20\,cm}) = \frac{2}{3}(60\,cm) = 40\,cm.$$

7. Calculate the displacement amplitude (in m) of a 40 kHz sound wave in helium if it has a pressure amplitude of 8×10^{-3} N/m^2. ($\rho = 0.179$ kg/m^3, $v = 972$ m/s.)

a. 1.8×10^{-10}
b. 3.7×10^{-10}
c. 7.8×10^{-9}
d. 2.4×10^{-9}
e. 1.9×10^{-10}

$f = 40 \times 10^3\ s^{-1}$
$\omega = 2\pi f$

$$\Delta P_{max} = \rho V \omega S_{max}$$

$$S_{max} = \frac{\Delta P_{max}}{\rho V \omega} = \frac{8 \times 10^{-3}\ \frac{N}{m^2}}{\left(0.179\ \frac{kg}{m^3}\right)\left(972\ \frac{m}{s}\right)\left(2\pi \times 40 \times 10^3\ s^{-1}\right)}$$

$$= 1.829 \times 10^{-4} \times 10^{-6} = \boxed{1.8 \times 10^{-10}\ m.}$$

$$\frac{N}{\frac{kg}{m^3} \cdot \frac{m}{s} s^{-1}} = \frac{\frac{kg \cdot m}{s^2}}{\frac{kg}{s^2}} = m$$

3

8. When you hear the horn of a car that is approaching you, the frequency that you hear is larger than that heard by a person in the car because

a. wave crests are farther apart by the distance the car travels in one period.
b. wave crests are closer together by the distance the car travels in one period.
c. the car gets ahead of each wave crest before it emits the next one.
d. the speed of sound in air is increased by the speed of the car.
e. a speeding car emits more wave crests in each period.

$y = A\cos\frac{\Delta\varphi}{2}\sin(kx - \omega t + \varphi)$ | $\frac{\Delta\varphi}{2} = 1.4 \Rightarrow \Delta\varphi = 2.8$

9. Two harmonic waves traveling in the same direction interfere to produce a travelling wave described by $y = 2\sin(\pi x - 3\pi t + 1.4)\cos(1.4)$ where x is in meters and t is in seconds. What is the difference of phase (in radians) between the two waves?

a. 0.7π b. 1.4 c. 0.7 d. 2.8 e. 1.4π

$V = 331\sqrt{1 + \frac{20}{273}} = 343\ \frac{m}{s}$ | $\frac{V}{4L} = \frac{343\ \frac{m}{s}}{1.20\ m} = 289$

10. A 30.0 cm tube in air (20°C) closed at one of its ends has a resonant frequency of 1429 Hz. Which harmonic is this?

a. 1 b. 2 c. 3 d. 4 e. 5

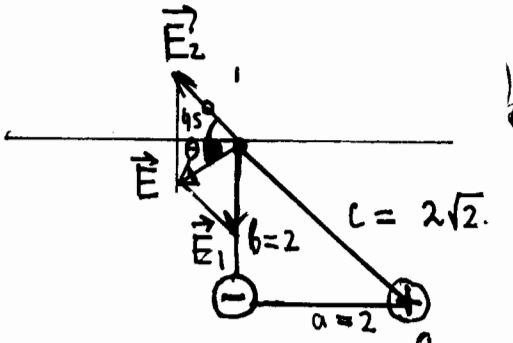
$f_n = n\cdot\frac{V}{4L}$

$1429 = (n)(289) \Rightarrow n \simeq 5$

11. A +20 nC point charge is placed on the x axis at $x = 2.0$ m, and a −25 nC point charge is placed at the origin of the xy- axes. What is the angle (measured in the counterclockwise direction from the positive x direction) that the direction of the resultant electric field at a point of coordinates (x=0 m, y=2.0 m) makes with the x-axis?

a. 209°
b. 61°
c. 29°
d. 248°
e. other (give the value)

$\theta_1 = 248°$
θ_1 is the angle which vector $\vec{E}$ makes with positive direction of x axis



$r_1 = b = 2.0\ m$
$r_2 = c = 2\sqrt{2}\ m.$

$\sin 45° = \cos 45° = \frac{1}{\sqrt{2}}.$

$q_1 = -25\ nC.\quad q_2 = +20\ nC.$

$$\vec{E} = \vec{E}_1 + \vec{E}_2 = k_e\frac{|q_1|}{r_1^2}(-\hat{j}) + k_e\frac{|q_2|}{r_2^2}(\sin 45°\hat{j} + \cos 45°(-\hat{i}))$$

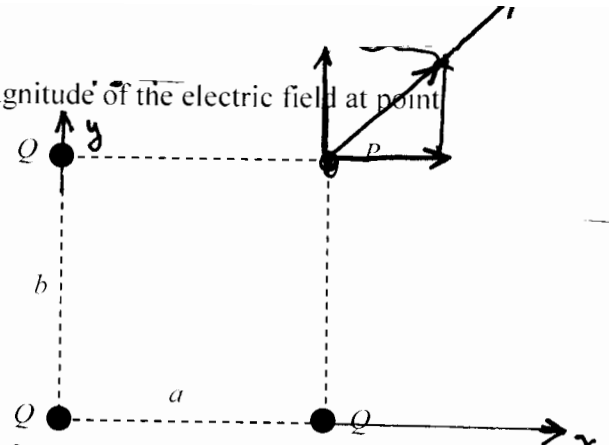
$$= k_e\left[\hat{i}\left(-\frac{|q_2|}{r_2^2}\frac{1}{\sqrt{2}}\right) + \hat{j}\left(\frac{|q_2|}{r_2^2}\frac{1}{\sqrt{2}} - \frac{|q_1|}{r_1^2}\right)\right] = E_x\hat{i} + E_y\hat{j}$$

$$= k_e\left[i\left(-\frac{20\ nC}{8\ m^2}\frac{1}{\sqrt{2}}\right) + j\left(\frac{20\ nC}{8\ m^2}\frac{1}{\sqrt{2}} - \frac{25\ nC}{4\ m^2}\right)\right]$$

$$\tan\theta = \frac{|E_y|}{|E_x|} = \frac{\frac{20\ nC}{8\ m^2}\frac{1}{\sqrt{2}} - \frac{25\ nC}{4\ m^2}}{\frac{20\ nC}{8\ m^2}\frac{1}{\sqrt{2}}} = 1 - \frac{25}{20}\frac{8}{4}\sqrt{2} = 2.536.$$

$\theta = \tan^{-1}(2.536) = 68° \Rightarrow \theta_1 = 180° + 68° = 248°$

12. If $Q = +10$ nC, $a = 3.0$ m, and $b = 3.0$ m, what is the magnitude of the electric field at point P?



a. 24 N/C

b. 31 N/C

c. 19 N/C

d. 15 N/C

e. 21 N/C

$$\vec{E} = k_e \frac{Q}{a^2}\hat{i} + k_e \frac{Q}{b^2}\hat{j} + k_e \frac{Q}{a^2+b^2}(\hat{i}\cos 45^\circ + \hat{j}\sin 45^\circ) =$$

$$= k_e Q\left\{\left[\frac{1}{a^2} + \frac{1}{\sqrt{2}(a^2+b^2)}\right]\hat{i} + \left[\frac{1}{b^2} + \frac{1}{\sqrt{2}(a^2+b^2)}\right]\hat{j}\right\} \overset{a=b}{=}$$

$$= \frac{k_e Q}{a^2}\left[\left(1+\frac{1}{2\sqrt{2}}\right)\hat{i} + \left(1+\frac{1}{2\sqrt{2}}\right)\hat{j}\right] = E_x\hat{i} + E_y\hat{j}$$

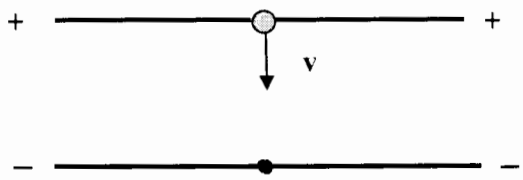
$$E_x = \frac{k_e Q}{a^2}\left(1+\frac{1}{2\sqrt{2}}\right); \quad E_y = \frac{k_e Q}{a^2}\left(1+\frac{1}{2\sqrt{2}}\right);$$

$$E^2 = \sqrt{E_x^2 + E_y^2} = \sqrt{\frac{k_e Q}{a^2}\left(1+\frac{1}{2\sqrt{2}}\right)^2 + \frac{k_e Q}{a}\left(1+\frac{1}{2\sqrt{2}}\right)^2} =$$

$$= \frac{k_e Q}{a^2}\left(1+\frac{1}{2\sqrt{2}}\right)\sqrt{2} = \frac{k_e Q}{a^2}\left(\frac{2\sqrt{2}+1}{2}\right) =$$

$$= \frac{8.988\times10^9 \frac{N^2\cdot m^2}{C^2}(10\times10^{-9}\,C)}{9.0\,m^2}\left(\frac{2\sqrt{2}+1}{2}\right) = 19\ N/C.$$

13. A proton enters the top plate of a parallel capacitor via a small opening and accelerates towards the bottom plate (see figure). The proton's entry speed is 1.20×10^4 m/s. The separation of the plates is 10 cm and the proton reaches the lower plate in 1.0 μs. What is the magnitude of the electric field in the capacitor?



a. 2.32×10^3 N/C

b. 2.06×10^3 N/C

c. 1.95×10^3 N/C

d. 1.84×10^3 N/C

e. other (give value)

$$d = v_i t + \frac{at^2}{2}, \quad a = \frac{qE}{m} \Rightarrow a = \frac{2(d - v_i t)}{t^2}$$

0.088 m

$$E = \frac{ma}{q} = \frac{2m(d - v_i t)}{q t^2} = \frac{2(1.6726\times10^{-27}\,kg)(0.1\,m - 1.2\times10^4\,\frac{m}{s}\times10^{-6}\,s)}{(1.6\times10^{-19}\,C)(10^{-6}\,s)^2}$$

$$= 0.184\times10^4\ \frac{N}{C} = 1.84\times10^3\ N/C.$$

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14. Two identical dust particles of mass 15.0×10^{-9} kg carry each a negative charge q. How many extra electrons does each dust particle contain (approximately), if the gravitational and electric forces between the particles balance each other?

a. 8 **b.** 27 **c.** 3.73×10^{3} **d.** 429 **e.** other (give the value)

b. 27 **c.** 3.73×10^{3} **d.** 429 **e.** other (give the value)

$$G\frac{m^2}{\cancel{r^2}} = k_e\frac{q^2}{\cancel{r^2}} \rightarrow q = \sqrt{\frac{Gm^2}{k_e}} = m\sqrt{\frac{G}{k_e}}$$

$$N = \frac{q}{|e|} = \frac{m}{|e|}\sqrt{\frac{G}{k_e}} = \frac{15.0\times10^{-9}\,kg}{1.6\times10^{-19}\,C}\sqrt{\frac{6.67\times10^{-11}\,\frac{N\cdot m^2}{kg^2}}{8.988\times10^{9}\,\frac{N\cdot m^2}{C^2}}} =$$

$$= \frac{15\sqrt{6.67}}{1.6\sqrt{8.988}}\times10^{-9+19-10} = 8.076 \simeq 8$$

15. Four identical -3.0 μC charges are located on the vertices of a square with 2.0 m long sides. What is the potential energy of an electron if placed at the center of the square?

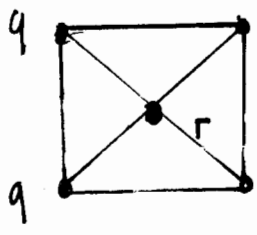
a. −38.1 keV

b. +38.1 keV

c. −76.5 keV

d. +76.5 keV

e. other (give value)



$a = 2.0$ m

$$r = \frac{1}{2}\sqrt{a^2+a^2} = \frac{1}{2}\sqrt{2^2+2^2} = \frac{1}{2}2\sqrt{2} = \sqrt{2}\ m.$$

$$V = \sum\frac{k_e q}{r} = 4\frac{k_e q}{r}\ ;\ U = qV = -|e|V$$

$$U = -\frac{|e|\cdot 4k_e(-|q|)}{r} = 4k_e\frac{|e||q|}{r} =$$

$$= \frac{4(8.988\times10^{9}\,\frac{N\cdot m^2}{C^2})(1.6\times10^{-19}C)(3.0\times10^{-6}C)}{\sqrt{2}\,m}\ \left(\frac{J}{C} = V\right) =$$

$$= \frac{4(8.988)(3.0)}{\sqrt{2}}\times10^{3}\left(\frac{N\cdot m^2}{C^2}\times\frac{C}{m}\right)\times(1.6\times10^{-19}C) =$$

$$= 76.3\ kV\times1.6\times10^{-19}\ C = 76.3\ keV$$

16. A particle (mass = 6.7×10^{-27} kg, charge = 3.2×10^{-19} C) moves along the positive x axis with a speed of 4.8×10^{5} m/s. It enters a region of uniform electric field parallel to its motion and comes to rest after moving 2.0 m into the field. What is the difference of potential between the entry point and the point it stops?

a. 2.4 kV b. 1.5 kV c. 3.5 kV d. 1.2 V e. 4.8 kV

$$\Delta(U+K) = \Delta U + \Delta K = 0 \Rightarrow \Delta U = -\Delta K.$$

$$\Delta V = \frac{\Delta U}{q} = \frac{-\Delta K}{q} = -\frac{1}{q}(K_f - K_i) = -\frac{1}{q}(0 - K_i) =$$

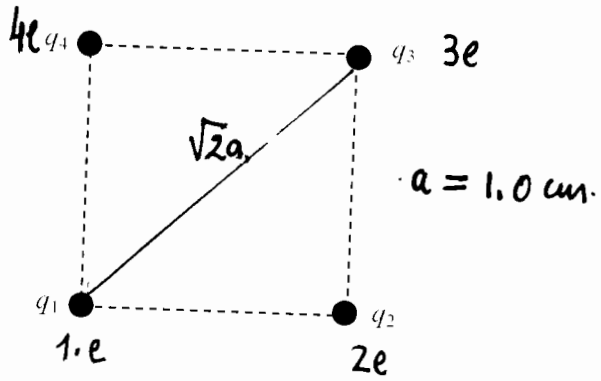
$$= \frac{K_i}{q} = \frac{m v_i^2}{2q} = \frac{6.7\times10^{-27}\,kg\,(4.8\times10^{5}\,\frac{m}{s})^2}{(3.2\times10^{-19}\,C)\,2} =$$

$$= \frac{6.7\,(4.8)^2}{3.2\,(2)}\,10^{2}\,\frac{kg\cdot m}{s^2}\,\frac{m}{C} = 24.12\times10^{2}\,\frac{J}{C} =$$

$$= 2.4\times10^{3}\ V = 2.4\ kV.$$

17. Four charges (q_1=1e, q_2=2e, q_3=3e, q_4=4e) are brought from infinity to the position shown below (see figure). Each charge occupies the corners of a square with edge length 1.0 cm. What is the potential energy of this configuration?

a. 3.0×10^{-5} eV
b. 4.6×10^{-6} eV
c. 8.7×10^{-2} eV
d. 6.2×10^{-7} eV
e. 9.1×10^{-10} eV



$$U = k_e\left[\frac{q_1q_2}{r_{12}} + \frac{q_1q_3}{r_{13}} + \frac{q_1q_4}{r_{14}} + \frac{q_2q_3}{r_{23}} + \frac{q_2q_4}{r_{24}} + \frac{q_3q_4}{r_{34}}\right]$$

$$= k_e\left[\frac{(1e)(2e)}{a} + \frac{(1e)(3e)}{\sqrt{2}a} + \frac{(1e)(4e)}{a} + \frac{(2e)(3e)}{a} + \frac{(2e)(4e)}{\sqrt{2}a} + \frac{(3e)(4e)}{a}\right]$$

$$= \frac{k_e e^2}{a}\left[2 + \frac{3}{\sqrt{2}} + 4 + 6 + \frac{8}{\sqrt{2}} + 12\right] = \frac{k_e e^2}{a}\left(24 + \frac{11}{\sqrt{2}}\right)$$

$$= \frac{8.988\times10^{9}\,\frac{N\cdot m^2}{C^2}\,(1.6\times10^{-19}\,C)\,(1.6\times10^{-19}\,C)}{1.0\times10^{-2}\,m}\,\overset{31.778}{\left(24 + \frac{11}{\sqrt{2}}\right)} =$$

$$= 8.988\,(1.6)\times10^{9-19+2}\times(31.778)\ V\times(1.6\times10^{-19}\,C) =$$

$$= 457\times10^{-8}\ V\ (1.6\times10^{-19}\,C) = 4.6\times10^{-6}\ eV$$

$$\frac{A}{C\,\frac{m}{s}\,m^2} = \frac{A}{C/s}\,\frac{1}{m^3} = \frac{1}{m^3} = m^{-3}.$$

18. When a current of 1.0 A flows in a wire with cross section area 4.0×10^{-6} m^2, the drift speed of the carriers is 4.7×10^{-5} m/s. What is the charge carriers' number density?

a. 4.3×10^{26} m^{-3}
b. 1.2×10^{27} m^{-3}
c. 4.5×10^{28} m^{-3}
d. 9.2×10^{26} m^{-3}
e. 3.3×10^{28} m^{-3}

$$I = nqv_dA \Rightarrow n = \frac{I}{q v_d A}$$

$$n = \frac{1.0\,A}{1.6\times10^{-19}C\,(4.7\times10^{-5}\,\frac{m}{s})(4.0\times10^{-6}\,m^2)}$$

$$= 0.033\times10^{30}\,m^{-3} = 3.3\times10^{28}\,m^{-3}.$$

19. A charged particle (mass = 2.0 mg, charge = $-6.0\ \mu$C) moves in the positive direction along the x axis with a velocity of 3.0 km/s. It then enters a region where a magnetic field $\mathbf{B} = (2.0\hat{i} + 3.0\hat{j} + 4.0\hat{k})$ mT exists. What is the acceleration of the particle in this region?

$\vec{F} = q\vec{v}\times\vec{B}$

$2.0\,mg = 2.0\times10^{-3}\,g = 2.0\times10^{-6}\,kg$

a. $(36\hat{j} - 27\hat{k})$ m/s^2
b. $(-36\hat{j} + 27\hat{k})$ m/s^2
c. $(-24\hat{j} + 18\hat{k})$ m/s^2
d. $(24\hat{j} - 18\hat{k})$ m/s^2
e. $(24\hat{j} - 27\hat{k})$ m/s^2

$\vec{B}$, $\vec{v}$

$$\vec{a} = \frac{\vec{F}}{m} = \frac{q}{m}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3\times10^3\frac{m}{s} & 0 & 0 \\ 2.0\,T & 3.0\,T & 4.0\,T \end{vmatrix} =$$

$$= \frac{-6.0\times10^{-6}\,C}{2.0\times10^{-6}\,kg}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3\times10^3\frac{m}{s} & 0 & 0 \\ 2.0\times10^{-3}T & 3.0\times10^{-3}T & 4.0\times10^{-3}T \end{vmatrix} =$$

$$= -3.0\frac{C}{kg}\left[-\hat{j}\,12.0\,\frac{m}{s}\cdot T + \hat{k}\,9.0\,\frac{m}{s}\cdot T\right]$$

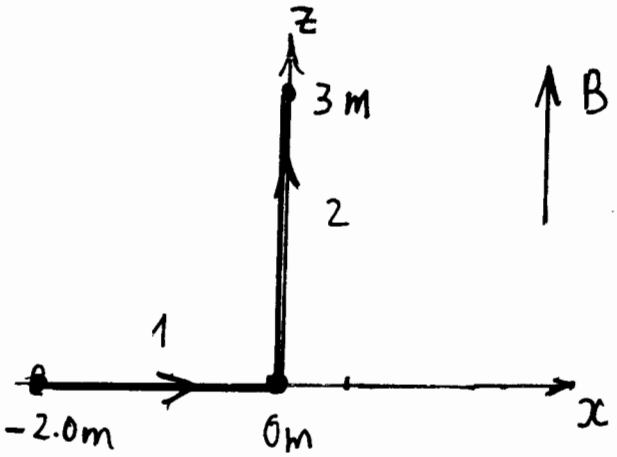
$$= +(36\hat{j} - 27\hat{k})\times10^3\,\frac{m}{s^2}.$$

$$\frac{C}{kg}\,\frac{m}{s}\,T = \frac{A\cdot m}{kg}\times\frac{N}{A\cdot m} = \frac{N}{kg} = \frac{\frac{kg\cdot m}{s^2}}{kg} = \frac{m}{s^2}.$$

$$1T = 1\,\frac{N}{A\cdot m}\ ;\ 1\,\frac{C}{s} = 1A$$

20. A segment of wire carries a current of 25 A along the x direction from $x = -2.0$ m to $x = 0$ and then along the z direction from $z = 0$ to $z = 3.0$ m. In this region of space the magnetic field is equal to 40 mT in the positive z direction. What is the magnitude of the total force on this segment of wire?

a. 1.0 N
b. 5.0 N
c. 2.0 N
d. 3.6 N
e. 3.0 N



$$\vec{F} = I\,\vec{l} \times \vec{B} = I l B \sin\theta$$

$$\vec{F} = \vec{F}_1 + \vec{F}_2 =$$

$$= I l_1 B \sin\frac{\pi}{2} + I l_2 B \sin 0 =$$

$$= I l_1 B = 25\,A\,(2\,m)\,(40 \times 10^{-3}\,T)$$

$$= 2000 \times 10^{-3}\,N = 2.0\,N$$

21. The magnetic field in a cyclotron of radius 1.20 m has a magnitude of 0.45 T. The maximum speed of the protons is achieved when the protons leave the cyclotron. What is this maximum speed?

a. 3.0×10^7 m/s
b. 8.5×10^8 m/s
c. 1.6×10^8 m/s
d. 3.2×10^8 m/s
e. 5.2×10^7 m/s

$$\frac{m v^2}{r} = q v B \Rightarrow \frac{m v}{r} = q B$$

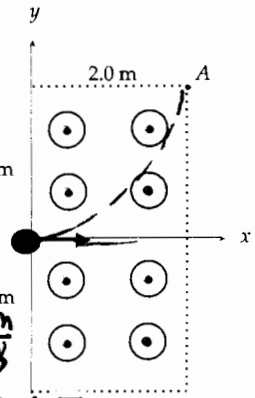
$$v = \frac{q B r}{m} = \frac{1.6 \times 10^{-19}\,C\,(0.45\,T)(1.20\,m)}{1.6726 \times 10^{-27}\,kg}$$

$$= 0.52 \times 10^8\,\frac{m}{s} = 5.2 \times 10^7\,\frac{m}{s}$$

$$\frac{C \cdot T \cdot m}{kg} = \frac{\cancel{A} \cdot s \cdot N \cdot \cancel{m}}{kg \cdot \cancel{A} \cdot \cancel{m}} = \frac{\frac{kg \cdot m}{s^2} \cdot s}{kg} = \frac{m}{s}$$

$$\frac{kg}{C\cdot s} = \frac{kg\cdot m}{\frac{C}{s}\, s^2\, m} = \frac{N}{A\cdot m} = 1T$$

22. The boundary shown in the figure is that of a uniform magnetic field directed in the positive z direction (out of the paper). An electron enters the magnetic field at the origin with a velocity in the positive x direction and exits 0.63 μs later at point A. What is the magnitude of the magnetic field?



a. 18 μT b. 34 μT c. 28 μT d. 14 μT e. 227 μT

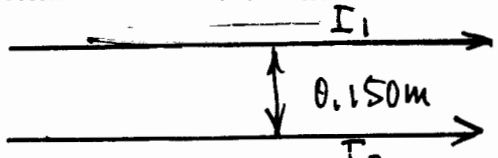
The electron goes around 1/4 of the circle $\Rightarrow$ $\tau = \frac{1}{4}T$

$$\frac{mv^2}{r} = qvB \rightarrow \frac{v}{r} = \frac{qB}{m} \text{ or } \frac{r}{v} = \frac{m}{qB} \Rightarrow T = \frac{2\pi r}{v} = \frac{2\pi m}{qB}$$

$$\tau = \frac{1}{4}\frac{2\pi m}{qB} \Rightarrow B = \frac{\pi m}{2q\tau} = \frac{\pi(9.1\times10^{-31}kg)}{2(1.6\times10^{-19}C)\,0.63\times10^{-6}s} = 14.2\times10^{-6}T = 14\,\mu T$$

23. Two long horizontal parallel wires are attracted to each other with a force per unit length of 320 μN/m when they are separated by a vertical distance of 0.150 m. The current in the upper wire is 20.0 A. Determine the current in the lower wire.

$$\frac{N}{T\cdot m} = \frac{A}{\frac{A}{A\cdot m}\cdot m} = A$$



a. 20.0 A
b. 40.0 A
c. 33.0 A
d. 12.0 A
e. 15.0A

$$F = \mu_0 \frac{I_1 I_2}{2\pi a} \Rightarrow I_2 = \frac{F\,2\pi a}{\mu_0 I_1}$$

$$I_2 = \frac{320\times10^{-6}\,N/m\,(2\pi)(0.150\,m)}{4\pi\times10^{-7}\,\frac{T\cdot m}{A}\,(20\,A)} = 12A$$

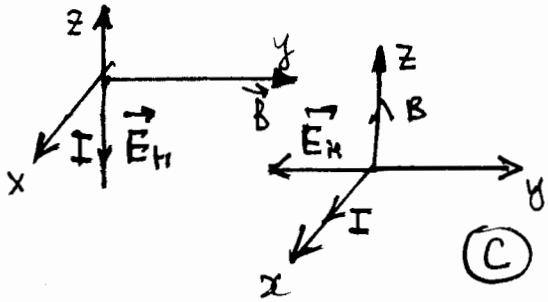
24. A metal strip has dimensions 7 cm × 4 cm × 2 cm in the $x\times y\times z$ directions respectively and is placed in a magnetic field. Which of the following statements is true?

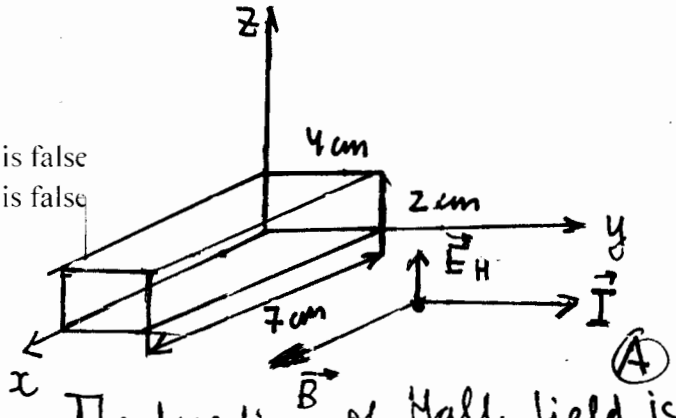
Statement A: If the current in the metal strip is in the $+y$ direction, the magnetic field is in the $+x$ direction and the Hall electric field is in the $+z$ direction, then the carriers are electrons.

Statement B: If the current in the metal strip is in the $+x$ direction, the magnetic field is in the $+y$ direction and the Hall electric field is in the $-z$ direction, then the carriers are electrons.

Statement C: If the current in the metal strip is in the $+x$ direction, the magnetic field is in the $+z$ direction and the Hall electric field is in the $-y$ direction, then the carriers are electrons.

a. only statement A is true
b. only statement B is true
c. only statement C is true
d. statements A and C are true but B is false
e. statements B and C are true but A is false





The direction of Hall field is opposite to the direction of $\vec{v}_d \times \vec{B}$

$\vec{v_d}$ is in $+y$ direction

A. If $q>0$ $\vec{v_d} \parallel \vec{I}$ and $\vec{v_d} \times \vec{B}$ is in $-z$ direction.

Then $\vec{E_H}$ is in $+z$ direction.

If $q<0$ $\vec{v_d}$ is $-y$ direction. $\vec{v_d} \times \vec{B}$ is in $+z$ direction.

and $\vec{E_H}$ is in $-z$ direction.

Since $\vec{E_H}$ is in $+z$ direction the carries are <u>not</u> electrons.

<u>Statement A is false</u>

$\vec{v_d}$ is in $+x$ direction

B. If $q>0$ $\vec{v_d} \parallel \vec{I}$ and $\vec{v_d} \times \vec{B}$ is in $+z$ direction.

Then $\vec{E_H}$ is in $-z$ direction.

If $q<0$ $\vec{v_d}$ is in $-x$ direction and $\vec{v_d} \times \vec{B}$ is in $-z$ direction. Then $\vec{E}$ is in $+z$ direction.

Since $\vec{E_H}$ is in $-z$ direction, the carries are <u>not</u> electrons

<u>Statement B is false.</u>

$\vec{v_d}$ is in $+x$ direction.

C. If $q>0$, $\vec{v_d} \parallel \vec{I}$ and $\vec{v_d} \times \vec{B}$ is in $-y$ direction.

Then $\vec{E_H}$ is y direction.

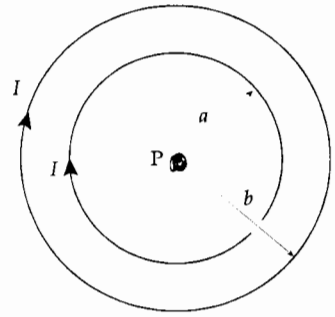
If $q<0$, $\vec{v_d}$ is in $-x$ direction and $\vec{v_d} \times \vec{B}$ is in y direction.

Then $\vec{E_H}$ is in $-y$ direction.

Since $\vec{E_H}$ is in $-y$ direction, the carries are electrons

<u>Statement C is true.</u>

25. What is the magnitude of the magnetic field at point P if $b = R$ and $a = R/2$?



a. $\frac{3\mu_0 I}{4R}$

b. $\frac{\mu_0 I}{4R}$

c. $\frac{3\mu_0 I}{2R}$

d. $\frac{\mu_0 I}{3R}$

e. $\frac{2\mu_0 I}{3R}$

Let $\hat{k}$ be unit vector out of the page.

$$\vec{B} = -\frac{\mu_0 I}{2a}\hat{k} - \frac{\mu_0 I}{2b}\hat{k} =$$

$$= -\frac{\mu_0 I}{2(R/2)}\hat{k} - \frac{\mu_0 I}{2R}\hat{k} =$$

$$= -\frac{2\mu_0 I}{2R}\hat{k} - \frac{\mu_0 I}{2R}\hat{k} = -\frac{3}{2}\frac{\mu_0 I}{R}\hat{k}$$

Magnitude of magnetic field is

$$B = \frac{3}{2}\frac{\mu_0 I}{2R}.$$

Since both currents are clockwise, the magnetic fields produced by the currents are directed into the page.

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$$\Delta\varphi = k\Delta r = \frac{2\pi}{\lambda}\Delta r = \frac{2\pi}{4m}\,5m = 2.5\pi =$$

$$= 2\pi + 0.5\pi = 2\pi + \frac{\pi}{2} = \varphi_2 - \varphi_1$$

If $y_1 = A\sin(kr - \omega t + \varphi_1)$, and

$$y_2 = A\sin(kr - \omega t + \varphi_2) =$$

$$= A\sin(kr - \omega t + \varphi_1 + \Delta\varphi).$$

Suppose that $y_1 = A\sin(kr_1 - \omega t + \varphi_1)$, and

$y_2 = A\sin(kr_2 - \omega t + \varphi_2)$.

$$\Delta\varphi = (kr_2 - \omega t + \varphi_2) - (kr_1 - \omega t - \varphi_1) =$$

$$= k(r_2 - r_1) + \varphi_2 - \varphi_1 = k\Delta r + (\varphi_2 - \varphi_1)$$

We assume that $\varphi_2 = \varphi_1$. Then $\Delta\varphi = k(r_2 - r_1) =$

$$= k\Delta r = \frac{2\pi}{\lambda}\Delta r = \frac{2\pi}{4m}\,5m = 2.5\pi = 2\pi + \frac{\pi}{2}.$$

If $y_1 = A\sin(kr_1 - \omega t)$

$$y_2 = A\sin(kr_1 + \Delta\varphi - \omega t) = A\sin(kr_1 \cdot$$

$$= A\sin(kr_1 - \omega t + 2\pi + \frac{\pi}{2})$$

$$kr_2 = kr_1 + \frac{\pi}{2} \Rightarrow \Delta\varphi = k(r_2 - r_1) = k\Delta r = \frac{\pi}{2}.$$

Answer: $\frac{\pi}{2}$ radians or $90°$.

$$n=0 : \; L = \frac{d^2 - \frac{1}{4}\left(\frac{v}{f}\right)^2}{\frac{v}{f}} = \frac{(4.00\,m)^2 - \frac{1}{4}\left(\frac{330\,\frac{m}{s}}{200\,s^{-1}}\right)^2}{\frac{330\,\frac{m}{s}}{200\,s^{-1}}} = 9.28\,m$$

$$n=1 : \; L = \frac{d^2 - \frac{9}{4}\left(\frac{v}{f}\right)^2}{\frac{v}{f}} = \frac{(4.00\,m)^2 - \frac{9}{4}\left(\frac{330\,\frac{m}{s}}{200\,s^{-1}}\right)^2}{\frac{330\,\frac{m}{s}}{200\,s^{-1}}} = 1.99\,m$$

$$n=0 : \quad L = \frac{d^2 - \frac{1}{4}\left(\frac{v}{f}\right)^2}{\frac{v}{f}} = \frac{(4.00\,m)^2 - \frac{1}{4}\left(\frac{330\,\frac{m}{s}}{200\,s^{-1}}\right)^2}{\frac{330\,\frac{m}{s}}{200\,s^{-1}}} = 9.28\,m$$

$$n=1 : \quad L = \frac{d^2 - \frac{9}{4}\left(\frac{v}{f}\right)^2}{\frac{v}{f}} = \frac{(4.00\,m)^2 - \frac{9}{4}\left(\frac{330\,\frac{m}{s}}{200\,s^{-1}}\right)^2}{\frac{330\,\frac{m}{s}}{200\,s^{-1}}} = 1.99\,m$$

Frequency of the oscillator can be calculated as

$$f = \frac{n}{2L}\sqrt{\frac{T_1}{\mu}} = \frac{n}{2L}\sqrt{\frac{m_1 g}{\mu}} = \frac{4}{2(2.00\,m)}\sqrt{\frac{25.0\,kg\,(9.8\,\frac{m}{s^2})}{0.00200\,\frac{kg}{m}}} = 350\,Hz$$

or as $f = \frac{(n+1)}{2L}\sqrt{\frac{T_2}{\mu}} = \frac{(n+1)}{2L}\sqrt{\frac{m_2 g}{\mu}} = \frac{5}{2(2.00\,m)}\sqrt{\frac{16.0\,kg\,(9.8\,\frac{m}{s^2})}{0.00200\,\frac{kg}{m}}} = 350\,Hz$

b. What is the largest mass for which standing wave can be observed?

$$f = \frac{n}{2L}\sqrt{\frac{T}{\mu}} = \frac{n}{2L}\sqrt{\frac{mg}{\mu}} \quad (n = 1, 2, 3, \ldots)$$

$$n \geq 1 \Rightarrow 2Lf\sqrt{\frac{\mu}{mg}} = n \geq 1$$

$$\sqrt{mg} \leq 2Lf\sqrt{\mu} \Rightarrow m \leq \frac{(2Lf)^2\mu}{g}$$

$$m_{max} = \frac{(2Lf)^2\mu}{g} = \frac{2^2(2.00\,m)^2(350\,s^{-1})^2(0.00200\,\frac{kg}{m})}{9.8\,\frac{m}{s^2}} = 400\,kg$$

The cylinder is a pipe open at the top and closed at the water surface.

The standing wave in m th normal mode has wavelength

$\lambda_m = \frac{4L}{m}, \quad m = 1, 3, 5, \ldots$

Resonance occurs each time the height of the air column changes by $\frac{\lambda(m+2)}{4} - \frac{\lambda m}{4} = \frac{\lambda}{2}$.

$\Delta L = \frac{\lambda}{2} = \frac{1}{2}\frac{v}{f}$.

The volume of the pipe between two water levels is

$\Delta V = \pi r^2 \cdot \frac{1}{2}\frac{v}{f}$, where r is the radius of the pipe.

$\Delta V = R\Delta t$, where R is the rate at which water is pumped into the pipe, Δt is time interval between successive resonances; $R\Delta t = \frac{\pi r^2 v}{2f} \rightarrow \Delta t = \frac{\pi r^2 v}{2fR}$

$\Delta R = 1\frac{L}{min} = \frac{1000 cm^3}{60 s} = \frac{0.001 m^3}{60 s}$

$$\Delta t = \frac{\pi (0.05 m)^2 (343 \frac{m}{s})}{2 (512 \frac{1}{s}) \frac{0.001 m^3}{60 s}} = 158 s$$